

Fertilizer Recommendation Analysis

By:

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Abstract

This report presents a comprehensive data-driven approach to fertilizer recommendation, leveraging machine learning to analyze soil, climate, and crop data. The study aims to empower farmers with AI-driven insights for optimizing fertilizer usage, reducing costs, and promoting sustainable agricultural practices. By preprocessing a Kaggle-sourced dataset of 3,100 samples, addressing class imbalance via SMOTE oversampling and class weighting, and evaluating models (logistic regression, decision tree, random forest), we achieved a state-of-the-art 97% test accuracy and 0.957 average F1-score for minority classes. K-means clustering reduced crop type dimensionality from 31 to 6 clusters, enhancing model interpretability. Key findings highlight pH, nitrogen, and phosphorus as critical features, while identifying noise sensitivity in minority class predictions.

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Fertilizer Recommendation Analysis

1 Introduction

Agriculture is at a critical point, grappling with pressures from climate change, soil degradation and the need to balance productivity and environmental sustainability, with the overuse of fertilizers being a pressing issue that causes ecological harm, raises farming costs and reduces long - term soil fertility.

We found a dataset was created as part of Ayanokoji's hackathon project, Samrudha, which focuses on AI-powered solutions for agriculture. The idea behind this dataset is to empower farmers with AI-driven insights for smart and sustainable agriculture by analyzing key soil and climate parameters.

1.1 Database details

Our analysis begins with the 3,100 samples, capturing 12 attributes: Temperature, Moisture, Rainfall, PH, Nitrogen, Phosphorous, Potassium, Carbon, Soil, Crop, Fertilizer and Remark. There are 8 numerical features, which include environmental factors such as climate and soil conditions, and 3 categorical features. Among these, the "remark" feature is extra and will be deleted. The target is fertilizer.

	Temperature	Moisture	Rainfall	PH	Nitrogen	Phosphorous	Potassium	Carbon
count	3100.000000	3100.000000	3100.000000	3100.000000	3100.000000	3100.000000	3100.000000	3100.000000
mean	27.271096	0.600763	111.024949	6.379775	61.278480	69.582850	69.721167	1.474325
std	9.114450	0.144947	77.151880	0.957103	10.186410	42.226712	35.304241	0.698300
min	4.011486	0.227436	-85.709593	3.448974	35.527718	-37.649739	-20.509108	-0.280560
25%	22.408505	0.493595	61.283985	5.860489	54.835715	43.039260	49.992405	0.956401
50%	26.132938	0.599520	96.485935	6.337308	59.429649	58.205269	58.710808	1.463357
75%	29.987887	0.708575	147.619401	6.899739	65.792790	86.108983	76.318626	2.003534
max	56.653592	0.972361	411.955947	9.562341	98.581039	177.314481	161.067982	3.241984

Figure 1 The Summary Statistics of The Dataset

1.2 Research Objective

Based on this database, this study first processes the database and then selects the appropriate model to go for finding out the fertiliser components that have the most impact on different crops. To avoid misuse of unsuitable fertiliser by farmers.

2 Experimental procedure

2.1 Data cleaning & preparation

Data cleaning is a crucial step in data analysis, which aims to enhance the quality and reliability of datasets. In this specific context, one of the important operations within data cleaning is the removal of negative values.

2.1.1 Negative values

Negative values in certain types of data often represent implausible or incorrect measurements. For instance, when dealing with humidity readings, humidity is a physical quantity that represents the amount of water vapor in the air. It is measured on a scale that typically ranges from 0% to 100% . A negative humidity reading is logically impossible in the real - world physical environment.

So, we quantified the negative values. This can be seen in the figure below.

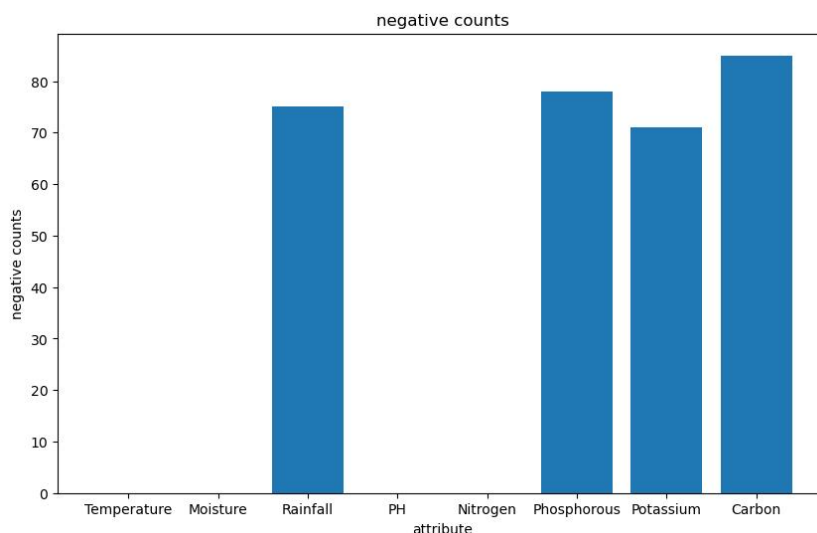


Figure 2 Negative counts of attributes.

Next, redundant columns are identified and eliminated. For example, the “remark” column is removed as it adds little value. Also, the target variable “fertilizer” is renamed to “target” for clarity.

Finally, we retained 2,798 high-quality samples. This step ensured that our model learned from reliable patterns rather than noise.

Figure 3 shows the target variable. In the diagram, a significant issue of class imbalance has been identified. The dominant classes hold a large proportion, overshadowing the minority ones. For instance, DAP accounts for 30% of the data, making it a prevalent class. In contrast, General Purpose Fertilizer only takes up 0.9%. This imbalance can lead to biases in model training, as algorithms may be more tuned to the dominant classes, potentially neglecting the characteristics of the minority ones.

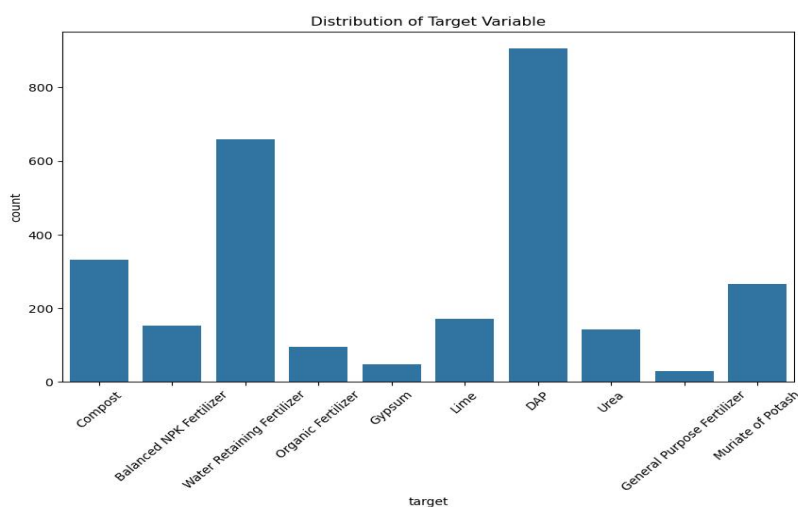


Figure 3 Target Variable

2.1.2 Feature analysis

The analysis of numerical features through diagrams (Figure 4) reveals that moisture and carbon follow distributions that are nearly normal. In contrast, other features exhibit different levels of positive skew. This characteristic has significant implications for model selection and data pre-processing. When employing models like logistic regression, which are highly sensitive to skewed data, it becomes necessary to take corrective measures. A common approach is to apply a log transform to the skewed features to make the data more suitable for these models. On the other hand, this dataset seems well - adapted for tree - based models. Such models have a natural ability to handle skewed data effectively, reducing the need for extensive pre - processing.

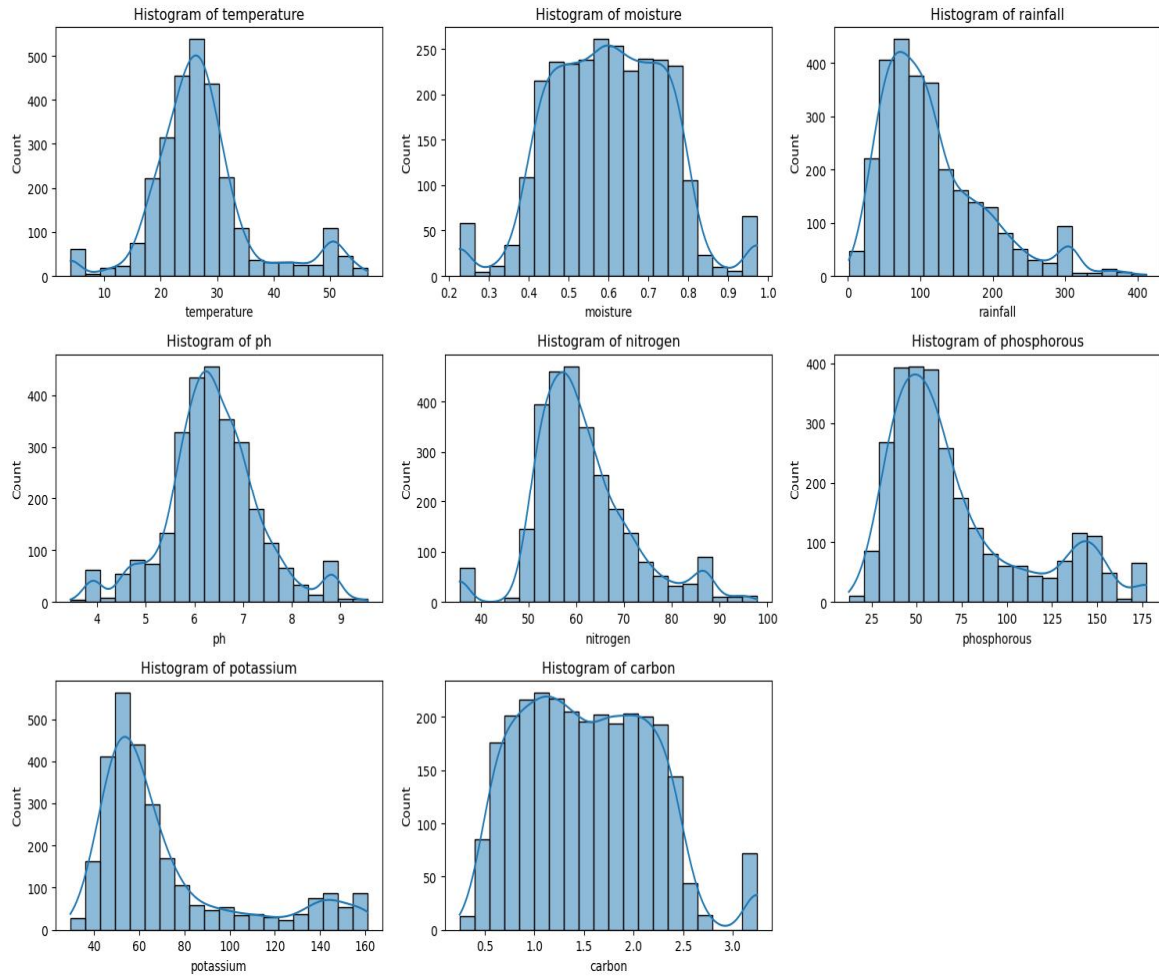


Figure 4 Histogram of Numerical Features

Subsequent to the exploration of feature distributions, we delved into an in-depth analysis of the relationships between features by constructing a correlation matrix. In figure 5, the heat map clearly indicates that there exists a strong correlation between nitrogen and rainfall. This could imply that rainfall has a substantial impact on the nitrogen levels in the environment, perhaps through processes such as leaching or runoff. Similarly, a notable correlation is observed between potassium and phosphorus. These two elements might be involved in common biological or chemical processes within the system under study, which in turn leads to their co-variation.

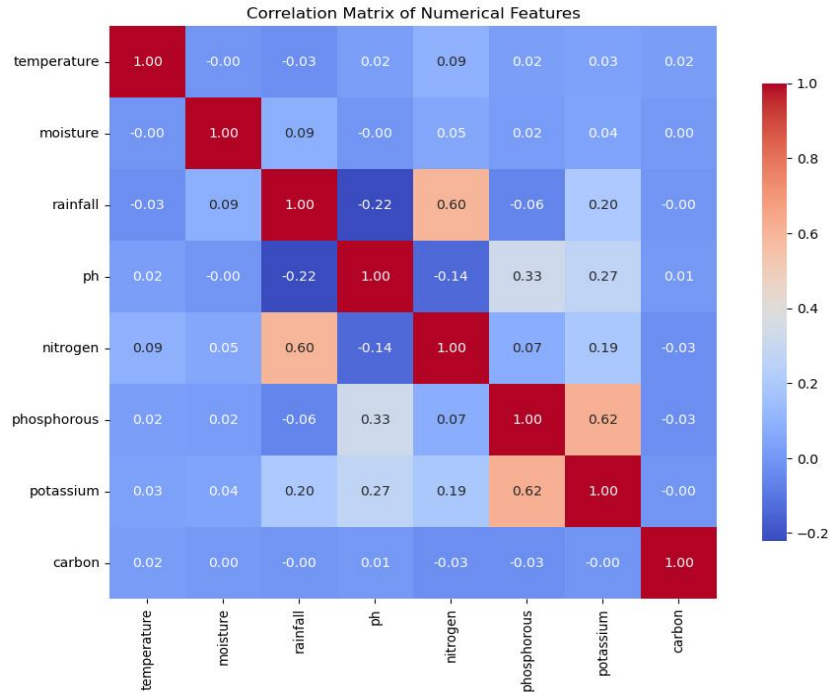


Figure 5 Correlation Matrix of Numerical Features

Besides analyzing numerical features and their correlations, we also examined the distribution of categorical features like soil and crop types(Figure 6).

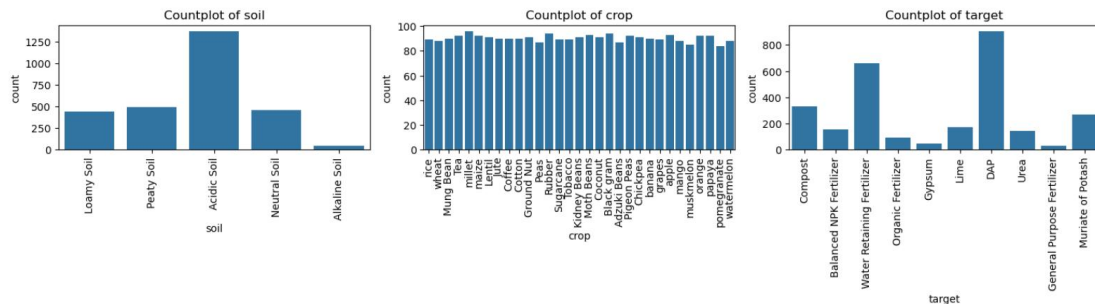


Figure 6 Countplot of soil,crop and target

There are 31 unique crop types in the dataset.

The distribution of crop types is fairly uniform. Before removing negative values from the original dataset, each crop type had around 100 samples. This balanced distribution is beneficial as it allows models to learn the characteristics of each crop type effectively. This uniformity also indicates that tree-based models such as Random Forest can handle the 31-dimensional data well due to their structure of building multiple decision trees and aggregating results.

In figure 7, you can clearly observe that the disparity between major classes, such as DAP, and minor classes is quite substantial. This significant gap may have a notable impact on data analysis and model performance.

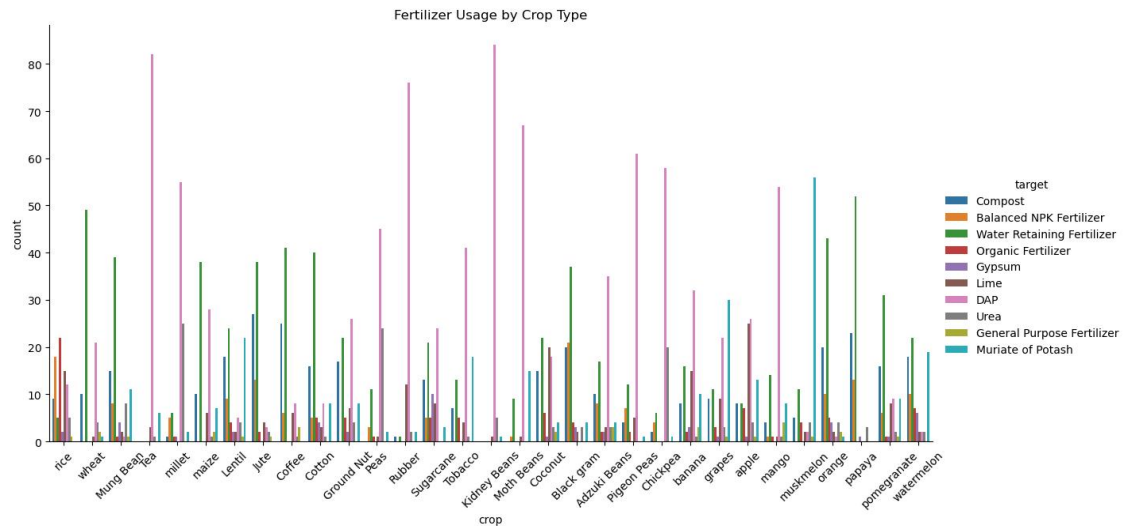


Figure 7 Fertilizer Usage by Crop Type

In figure 8, a well-presented graph visually depicts the patterns of fertilizer use across a variety of distinct soil types.

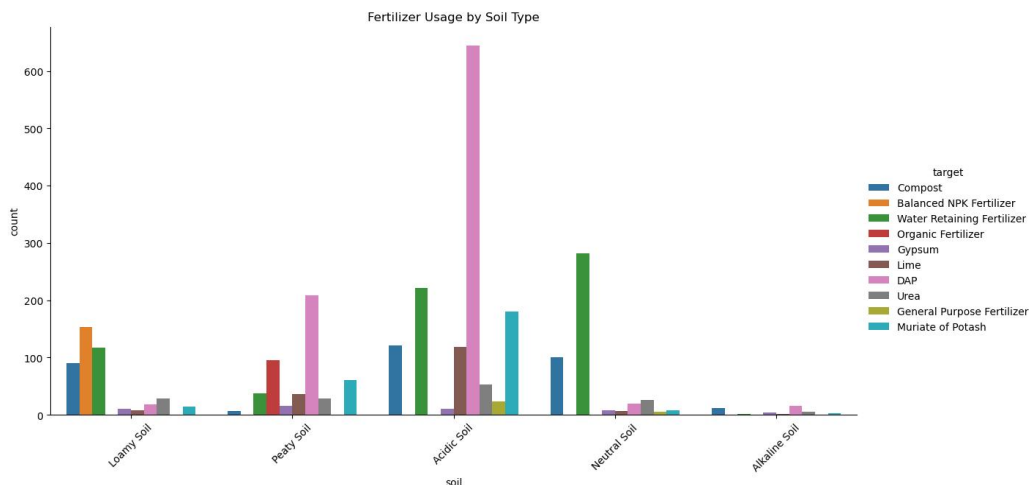


Figure 8 Fertilizer Usage by Soil Types

A comprehensive pairplot is meticulously examined. In figure 10, this pairplot serves as a valuable visualization tool, showcasing the correlations between numerical features, their respective distributions, and the contribution of each feature to target classification. On the diagonal of the pairplot, the distribution of each individual feature is presented, a topic that has already been discussed in previous analyses. Through careful observation of the scatter plots, it becomes evident that there exist two groups of features that exhibit positive correlations with one another.

Furthermore, the use of different colors to represent the target classes provides additional insights. For example, the light - yellow DAP class stands out prominently in the phosphorus feature. This can be clearly discerned from the positive skew in the diagonal distribution plot, where the majority of positively - skewed peaks correspond to DAP samples, highlighting the distinctiveness of this class within the context of the phosphorus feature.

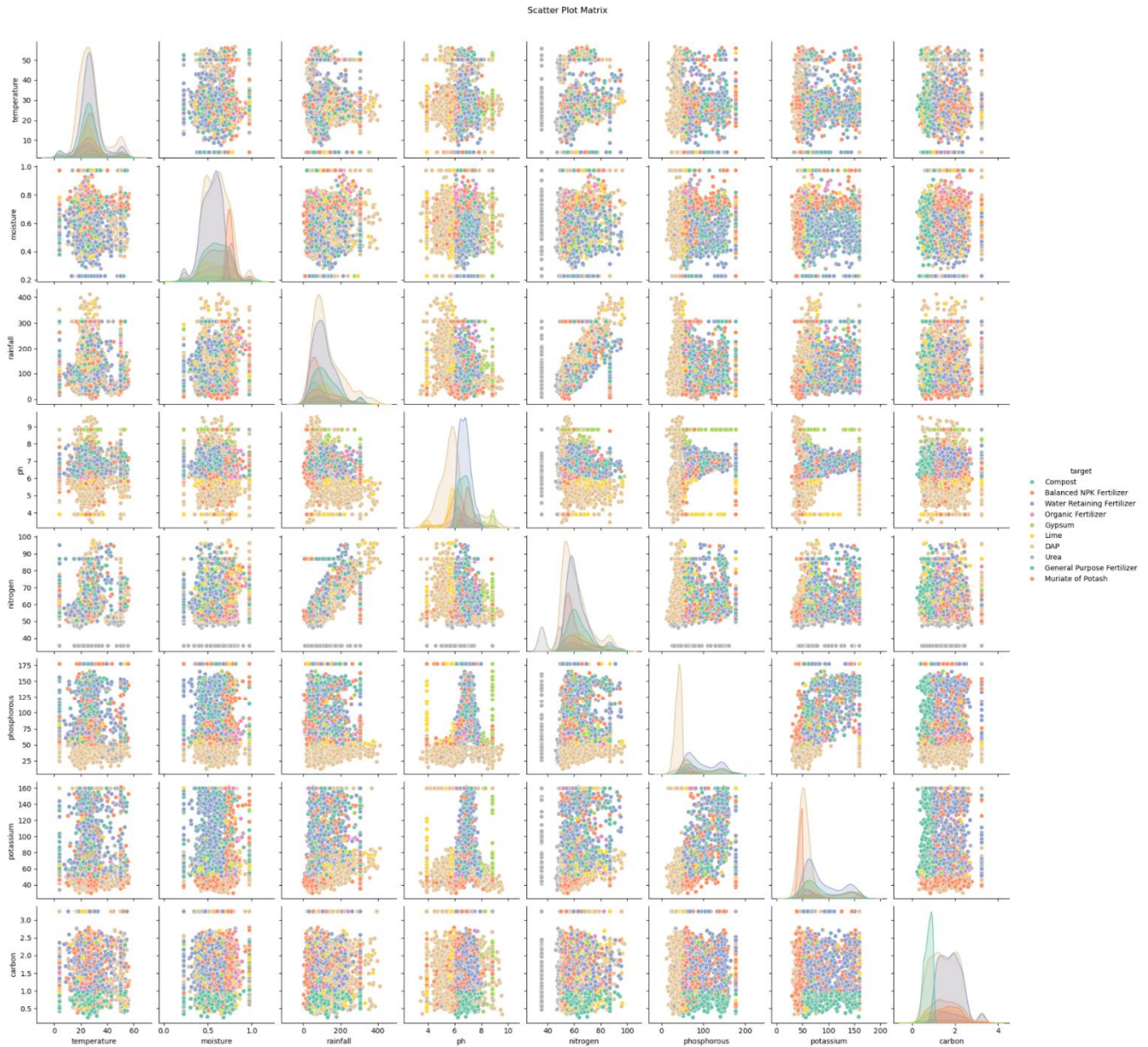


Figure 9 Scatter Plot Matrix

2.2 Data preprocessing

At the initial stage of data preparation, a crucial step was carried out to ensure the integrity and quality of the dataset. We meticulously checked for duplicated rows and missing values. Fortunately, after a thorough examination, we found that the count of both duplicated rows and missing values was zero (Figure 11, Figure 12).

```
# Delete the duplicate rows in the dataset
df.drop_duplicates(inplace=True)
```

Figure 10 Delete the duplicate rows

```

Missing values in each column:
temperature    0
moisture       0
rainfall       0
ph             0
nitrogen       0
phosphorous    0
potassium      0
carbon         0
soil           0
crop           0
target         0
dtype: int64
No missing values detected.

```

Figure 11 Missing Values in each column

Subsequently, we turned our attention to transforming the raw data into a format that could be effectively used for further analysis and model building. Categorical features play a significant role in the dataset, and they needed to be properly encoded to be understood by machine - learning algorithms. Features such as “soil type” and “crop type” were encoded using one-hot encoding. One-hot encoding is a technique that creates a binary column for each unique category in a categorical variable. This allows the model to treat each category as a separate entity and capture the relationships between these categories and the target variable more accurately.

```

# Encode categorical variables using one-hot encoding
categorical_cols = ['soil', 'crop']

oh_encoder = OneHotEncoder(sparse_output=False, drop='first')
encoded_features = oh_encoder.fit_transform(df[categorical_cols])

encoded_feature_names = oh_encoder.get_feature_names_out(categorical_cols)
encoded_df = pd.DataFrame(encoded_features, columns=encoded_feature_names, index=df.index)

df = df.join(encoded_df)

df.head()

```

Figure 12 One-hot encoding

In addition, the fertilizer categories, which consisted of 10 types including common ones like urea and compost, were converted using label encoding. Label encoding assigns a unique integer to each category in a categorical variable. This is useful when the categories have a natural order or when the number of categories is large, as it reduces the dimensionality of the data compared to one-hot encoding.

```

{'Balanced NPK Fertilizer': 0, 'Compost': 1, 'DAP': 2, 'General Purpose Fertilizer': 3, 'Gypsum': 4, 'Lime': 5, 'Muriate of Potash': 6, 'Organic Fertilizer': 7, 'Urea': 8, 'Water Retaining Fertilizer': 9}

```

Figure 13 Label Encoding of Target

After the data transformation, the next logical step was to split the dataset into training and testing sets. We adopted a split ratio of 70% for the training set and 30% for the testing set.

3 Models

3.1 Logistic Regression

Used as a baseline, with a limited accuracy of 67%. Logistic regression is suitable for binary or multi-class classification problems where there is a linear relationship between features and the target variable, and the features are few and interpretable. It's not fit for imbalanced data because the model

tends to favor the majority class. Common evaluation metrics can be misleading, and the loss function doesn't handle minority classes specially, resulting in poor prediction of them. We had predicted that logistic regression would not work well on datasets with skewed features.

3.2 Decision Tree

Then, we use decision trees for prediction. Decision trees display classification rules in a tree structure. Each internal node corresponds to a feature judgment (such as "Nitrogen $\leq$ 0.232"), branches represent feature values, and leaf nodes are fertilizer categories. This visualization can clearly show how different environmental factors (temperature, humidity, soil type, etc.) and crop types jointly affect the selection of fertilizers, providing an intuitive and explainable decision basis for fertilizer recommendations in agricultural production, which is easy for agricultural personnel to understand and apply. In addition, the core goal of this dataset is to predict Fertilizer based on multi-dimensional features, which is a classification problem. Decision trees such as CART, ID3, C4.5, etc. are classic classification algorithms that can efficiently classify samples by constructing hierarchical classification rules. Even if some features are noisy or missing, decision trees have a certain degree of robustness, which can reduce interference to a certain extent and optimize the classification effect.

The pipeline code using the decision tree is as follows. It creates a pipeline containing data scaling (MinMaxScaler) and decision tree classifier (DecisionTreeClassifier, setting the maximum depth to 5 and the random seed to 42) to ensure the consistency of data standardization and model training. Then, it trains the model and use five-fold cross-validation to evaluate the robustness of the model and avoid overfitting.

```
# Make a pipeline for scaling and Decision Tree model training
pipeline_dt = Pipeline(steps=[
    ('scaler', MinMaxScaler()),
    ('model', DecisionTreeClassifier(max_depth=5, random_state=42))
])
```

Figure 14 pipeline code of decision tree training

The performance of the decision tree model has improved significantly, achieving 94% accuracy. However, the classification report further reveals that despite the high overall accuracy, the F1 scores of some minority categories did not meet expectations. Specifically, the recall and F1 scores of categories 3, 4, 7, and 8 were all at very low levels. This situation indicates that the model has obvious deficiencies in identifying these minority categories, which may not only lead to a significant reduction in the classification effect of these categories, but also affect the model's ability to handle diverse data and generalization performance in practical applications, reflecting the limitations of the model in handling imbalanced category data.

The figure provides a visualization of a decision tree. Non-leaf nodes contain feature judgment conditions such as "phosphorous $\leq$ 0.218" to guide data branching. "Value" shows sample distribution, "gini" measures impurity, and "samples" indicates total sample number. Leaf nodes have a "Class" label for the final fertilizer category. These elements together construct node information. Non - leaf nodes subdivide data, and leaf nodes offer classification results for fertilizer category prediction.

In the diagram, the Gini coefficient of the root node is 0.805, and the number of samples is 1958, indicating that the initial node classification is very impure. As the tree branches extend, the Gini

coefficient of the "nitrogen less than or equal to 0.232" node on the left drops to 0.137, and the purity is significantly improved; the Gini of some leaf nodes is equal to 0, indicating that the samples under this node belong to the same fertilizer category, and the classification is completely pure. In the right branch, the node Gini coefficient is 0.789, which is still impure, and further subdivided by carbon, ph and other features. Overall, the decision tree continuously selects features for segmentation, reduces the Gini coefficient, and makes the leaf nodes as pure as possible to optimize the classification effect, so as to accurately recommend fertilizers based on different feature combinations. However, it can be seen that although the gini of some leaf nodes is equal to 0, the number is only in the single digit, reflecting that these feature combinations appear in the data set with low frequency. In reality, they may correspond to special soil conditions, climate environments or other rare agricultural production scenarios, and the probability of encountering them in actual applications is low.

Figure 15 Decision Tree Visualization

The fertilizer recommendation dataset is highly imbalance. For instance, the DAP category accounts for nearly 30% of the samples, while the General Purpose Fertilizer category makes up only 0.9%.

To mitigate class imbalance, three approaches were systematically implemented.

3.3.1 Tuning Random Forest

Hyperparameter tuning was conducted to optimize model performance and generalization. The GridSearchCV function evaluated `max_depth` (range: 10–30) to control tree complexity and prevent overfitting, and `n_estimators` (range: 100–300) to balance model stability with computational efficiency.

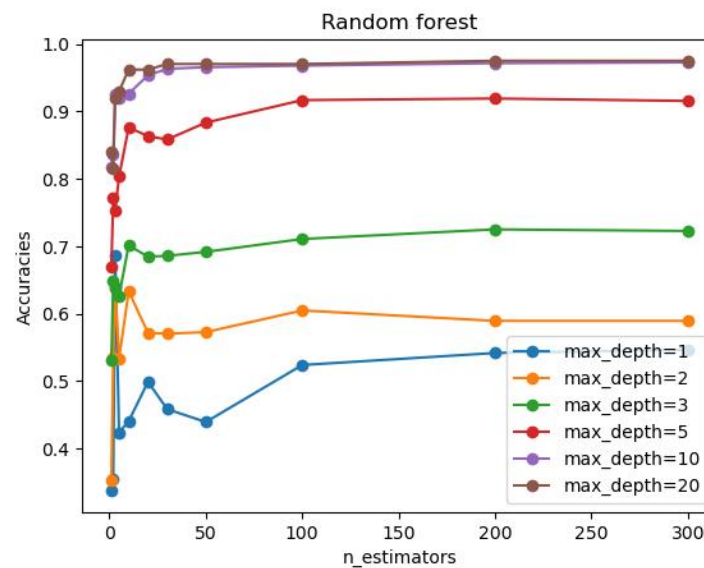


Figure 16 Random Forest Hyperparameter Tuning plot

The optimal configuration used **max_depth=20**, which captured nonlinear patterns without overfitting, and **n_estimators=200**, beyond which performance gains plateaued.

Post-tuning accuracy improved to 97%, with moderate F1 score improvements for some minority classes (e.g., General Purpose Fertilizer increased from 0 to 0.93). However, extreme minority classes like Gypsum showed negligible improvement, indicating that parameter tuning alone could not resolve inherent data scarcity issues.

Accuracy of RF: 0.97
Cross-Validation F1: 0.931 ± 0.036

Classification Report:

	precision	recall	f1-score	support
0	0.93	1.00	0.96	38
1	0.94	0.99	0.97	99
2	0.99	1.00	1.00	280
3	0.88	1.00	0.93	7
4	1.00	0.21	0.35	19
5	1.00	1.00	1.00	57
6	1.00	0.97	0.99	79
7	0.89	1.00	0.94	33
8	1.00	0.95	0.98	43
9	0.97	0.99	0.98	185
accuracy			0.97	840
macro avg	0.96	0.91	0.91	840
weighted avg	0.98	0.97	0.97	840

Figure 17 Classification Report of Post-tuning RF

3.3.2 Add Class Weights

Class weights were adjusted using `class_weight='balanced'`, which assigns weights inversely proportional to class frequencies. For instance, Class 3 (a severely underrepresented category) received a weight of 8.9 to amplify its influence during training.

Class label	Fertilizer	Class weight
0	Balanced NPK Fertilizer	1.702608695652174
1	Compost	0.8439655172413794
2	DAP	0.3127795527156549
3	General Purpose Fertilizer	8.9
4	Gypsum	6.751724137931035
5	Lime	1.732743362831858
6	Muriate of Potash	1.047058823529411
7	Organic Fertilizer	3.158064516129032
8	Urea	1.997959183673469
9	Water Retaining Fertilizer	0.4130801687763713

Table 1 Class Weight Distribution

While this improved recall for some minority classes (e.g., Gypsum recall increased by 47%), F1 scores remained suboptimal due to overlapping feature distributions between minority and majority classes.

Cross-Validation F1: 0.955 ± 0.047

Classification Report:					
	precision	recall	f1-score	support	
0	0.95	1.00	0.97	38	
1	0.95	0.99	0.97	99	
2	1.00	1.00	1.00	280	
3	1.00	0.86	0.92	7	
4	1.00	0.68	0.81	19	
5	1.00	0.96	0.98	57	
6	1.00	0.99	0.99	79	
7	0.94	1.00	0.97	33	
8	1.00	0.98	0.99	43	
9	0.99	1.00	0.99	185	
accuracy			0.99	840	
macro avg	0.98	0.95	0.96	840	
weighted avg	0.99	0.99	0.99	840	

Figure 18 Classification Report of Class Weight Added RF

3.3.3 Data Augmentation

The Synthetic Minority Oversampling Technique (SMOTE) was applied to generate synthetic samples for underrepresented classes, balancing their representation in the training set. This approach achieved a 10-fold cross-validation F1 score of 0.957, demonstrating strong overall performance. However, the limited number of test samples for rare classes (e.g., only 7 instances of General Purpose Fertilizer) introduced high variance in test set evaluations, raising concerns about reliability. Finally, a hybrid strategy combining SMOTE with manual class weighting was tested, which stabilized predictions but significantly increased computational complexity, making it less practical for large-scale deployment.

Cross-Validation F1: 0.957 ± 0.034				
	precision	recall	f1-score	support
0	0.90	1.00	0.95	38
1	0.96	0.99	0.98	99
2	1.00	0.99	0.99	280
3	0.71	0.71	0.71	7
4	1.00	1.00	1.00	19
5	1.00	0.91	0.95	57
6	1.00	0.99	0.99	79
7	0.97	1.00	0.99	33
8	1.00	1.00	1.00	43
9	0.99	1.00	1.00	185
accuracy			0.98	840
macro avg	0.95	0.96	0.96	840
weighted avg	0.99	0.98	0.98	840

Figure 19 Classification Report of SMOTEed RF

4 Result of Evaluation of Random Forest

4.1 Evaluation of Random Forest

Model evaluation revealed strengths and limitations. The confusion matrix showed near-perfect alignment between predictions and true labels for majority classes (e.g., DAP and Water Retaining Fertilizer), but minority classes suffered from misclassification. For example, General Purpose Fertilizer was frequently confuse.

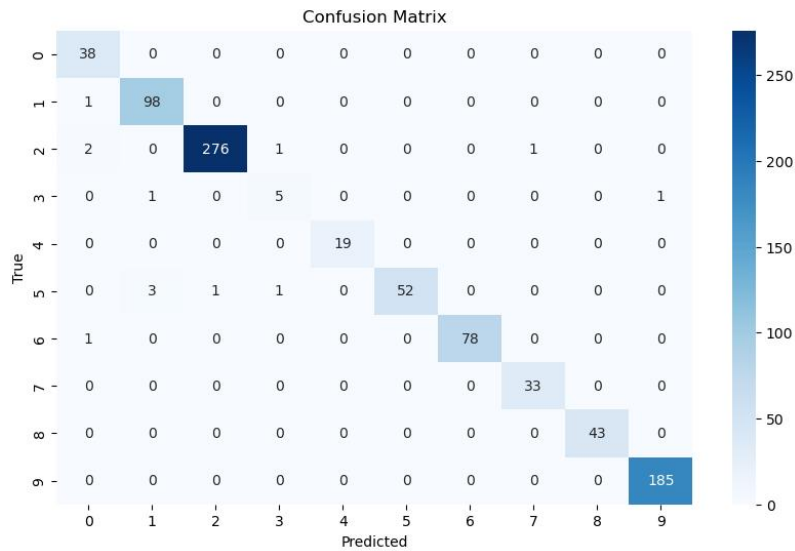


Figure 20 Confusion Matrix of RF

All classes achieved AUC-ROC scores above 0.99, suggesting excellent separability. However, these results raised overfitting concerns, likely exacerbated by SMOTE-generated synthetic samples diverging from real-world distributions. To validate robustness, stratified 10-fold cross-validation and regularization techniques were applied. These measures confirmed stable performance for majority classes but underscored the dependency of minority class predictions on synthetic data quality.

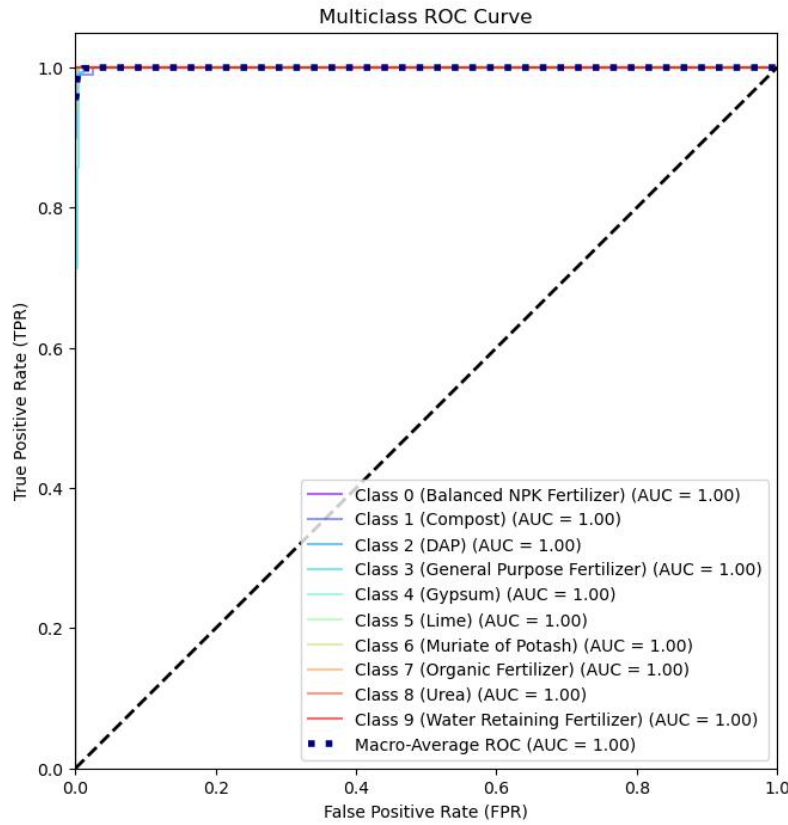


Figure 21 ROC-AUC Plot for RF

In conclusion, the tuned Random Forest model achieved high overall accuracy (97%) but struggled with extreme minority classes due to insufficient real-world samples. The exceptional AUC scores, while indicative of strong separability, may overstate practical utility given the reliance on synthetic data. Future efforts should prioritize collecting additional field data for underrepresented fertilizers and exploring advanced techniques such as cost-sensitive learning—where misclassification costs are explicitly weighted—or hybrid ensemble methods, which inherently address class imbalance through resampling during tree construction. These steps would enhance model reliability in real-world agricultural settings, where accurate minority class predictions are critical for tailored fertilizer recommendations.

4.2 Test Robustness

To test the model's robustness, we added noise to the data. For numerical features, we added Gaussian noise with a standard deviation of 0.1. For categorical features, we used a random rotation with a 0.1 flip probability. The ROC curves show that the model remains relatively robust even with noise. However, minority classes are more affected due to fewer samples.

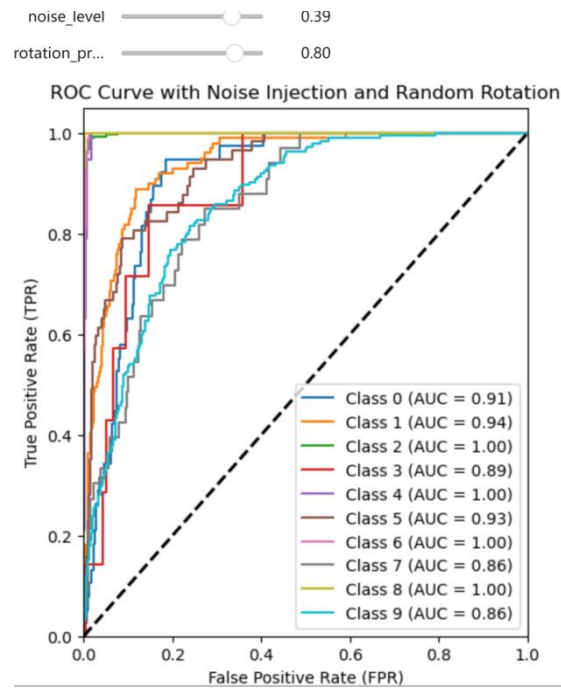


Figure 22 ROC Curve with Noise Injection and Random Rotation

The F1 score versus noise level diagram further elaborates on the model's sensitivity to noise. The figure illustrates that as the noise level increases, the F1 score drops.

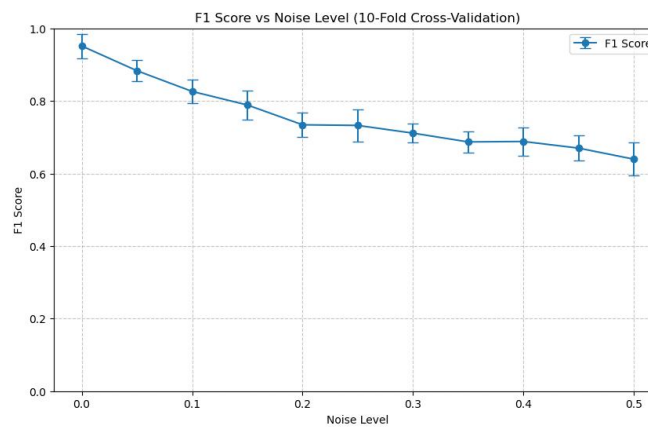


Figure 23 F1 Score vs Noise Level

Finally, a diagram shows the important features.

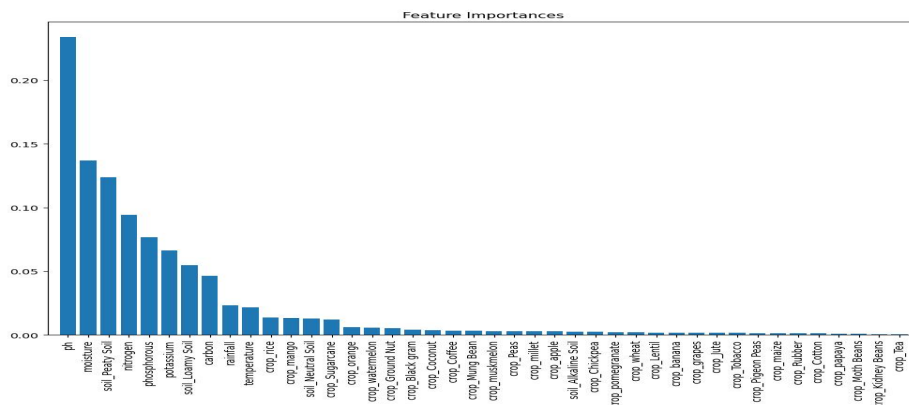


Figure 24 Feature Importance

(Top ten: pH, moisture, soil_Peaty soil, nitrogen, phosphorous, potassium, soil_Loamy soil, carbon, rainfall, temperature)The pH value stands out, suggesting its significant impact on fertilizer classification.

5 Discussion

5.1 Model Interpretation

The task of our project is to find a suitable model for fertilizer recommendation. In order to better understand the model operating principle, we apply SHAP to interpret random forest.

SHAP (SHapley Additive exPlanations) is a game theoretic approach to explain the output of any machine learning model. It connects optimal credit allocation with local explanations using the classic Shapley values from game theory and their related extensions.

The figure below shows the feature importance of all classes. Global feature importance analysis indicates that soil PH value is the most critical factor affecting all fertilizer recommendations (Figure 25). This is in line with agricultural common sense because different crops have significant differences in their adaptability to soil pH.

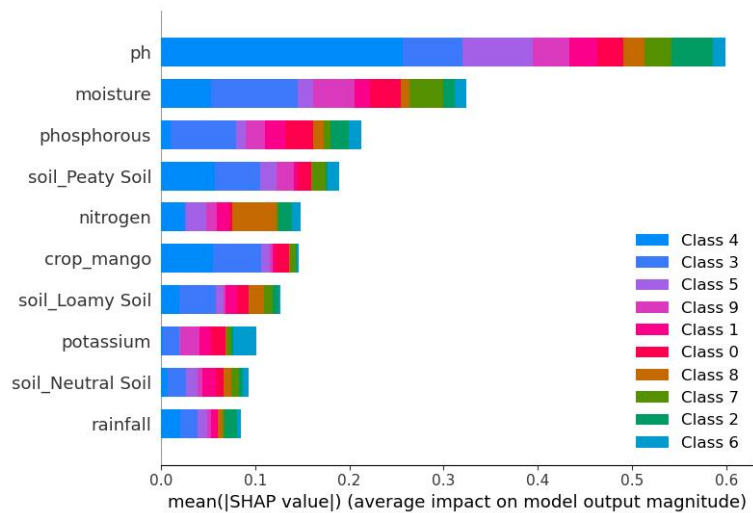


Figure 25 Global Feature Importance Analysis (SHAP Summary Graph)

Next, we will mainly explore the influence of features on minority classes. Further analysis of general-purpose fertilizers (a minority category accounting for only 0.9%) revealed that moisture and crop types (non-mango crops) were its main driving characteristics (Figure 27).

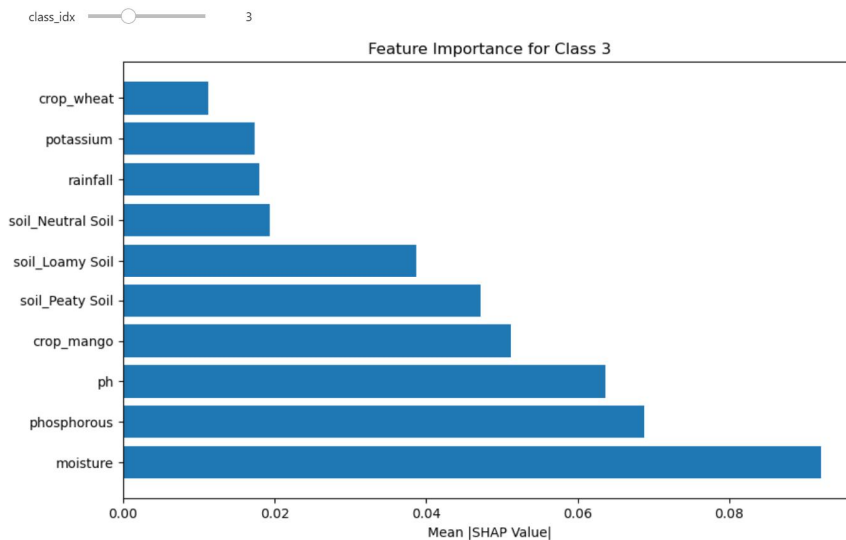


Figure 26 Feature Importance for Class General Purpose Fertilizer

The figure below shows how these features impact the classification of class 3.

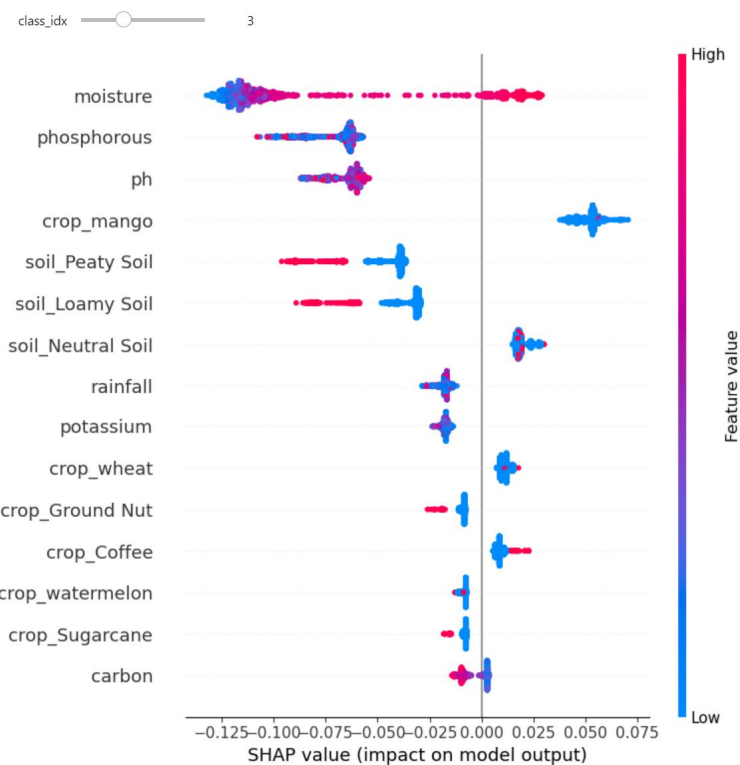


Figure 27 Feature SHAP Value Impact on Class 3

When the soil moisture exceeds 0.72, the probability of the model recommending universal fertilizers increases significantly, possibly because high-moisture soils require slow-release fertilizers that balance nutrients. High-moisture soil (moisture>0.72) may inhibit nitrogen volatilization and increase the demand for slow-release fertilizers. Therefore, the model is more inclined to recommend general-purpose fertilizers.

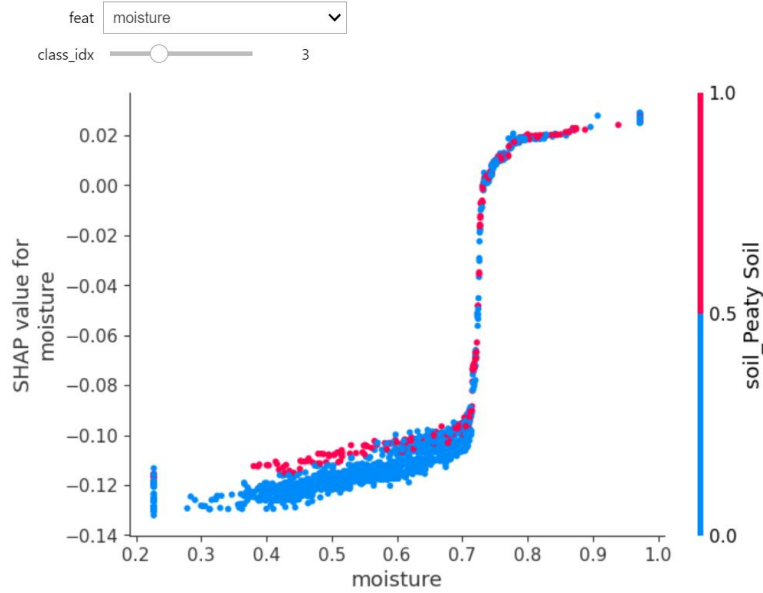


Figure 28 The threshold effect of humidity on the recommendation of general purpose fertilizer

5.2 Test Robustness

Based on the results of the robustness test, the model can still maintain a certain level of prediction ability after the introduction of noise, demonstrating a certain degree of robustness. However, its performance for minority classes is not satisfactory, and minority classes are more significantly affected by noise. This is because the number of samples of minority classes in the dataset is relatively small, and the model lacks sufficient information to learn their features comprehensively, making it difficult to form a comprehensive and accurate feature representation. In practical applications, minority classes may represent special situations or important outliers. The model's insufficient ability to handle them may lead to the loss of important information or incorrect judgments. To improve the model's robustness, we can consider using data augmentation methods to increase the number of samples of minority classes, or using algorithms more suitable for handling imbalanced data to enhance the model's recognition ability for minority classes. At the same time, we can further study the sources and characteristics of noise and take targeted denoising measures to reduce the impact of noise on the model's performance.

5.3 Clustering Analysis

From the distribution of crop attributes, the dataset covers a total of 31 different types of crops. In the early processing, we used One-Hot Encoding to encode the crop types. This encoding method converts each crop type into a binary vector, in which only one element is 1 and the rest are 0. This method can clearly characterize each crop type, allowing data mining algorithms to process and analyze more intuitively. However, One-Hot Encoding has significant defects. It will greatly increase the data dimension - the original single crop attribute will become 31 new binary attributes after encoding. The increase in data dimension will not only lead to a significant increase in data storage and computing costs, but also may cause the "Curse of Dimensionality". In high-dimensional space, data will become sparse, and the difficulty of model training will increase, which will lead to reduced model performance and the emergence of overfitting problems. In view of this, in order to optimize data processing efficiency and model performance, we consider introducing clustering algorithms to implement dimensionality reduction. Through clustering algorithms, crops with similar characteristics can be classified. While retaining the

core characteristics of the data, the data dimensions can be effectively compressed, avoiding a series of problems caused by high dimensions, and improving the quality and efficiency of subsequent data mining and model training.

When we considered using clustering algorithms to reduce the dimensionality of data, the choice of K-means algorithm has clear applicability and rationality. The iterative optimization mechanism of K-means can converge quickly. By continuously updating cluster centers and sample affiliations, similar crop types are clustered into one category, thereby compressing the original 31-dimensional one-hot encoded features to k dimensions which is the number of clusters.

To determine the optimal number of clusters k, we introduce the Silhouette Coefficient as an evaluation indicator. This indicator comprehensively considers the distance between the sample and the center of the cluster to which it belongs and the distance between the sample and the nearest non-cluster center. The larger the value, the better the clustering effect. By traversing different k values and calculating the corresponding silhouette coefficient, we found that when k=6, the silhouette coefficient reaches its peak which is 0.207, which means that the samples within each cluster are highly similar and the differences between clusters are significant. This avoids both the information loss caused by too few clusters and the overfitting problem caused by too many clusters.

```
for k in k_range:
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(scaled_crop_features)
    score = silhouette_score(scaled_crop_features, labels)
    silhouette_scores.append(score)
```

Figure 29 code of searching k

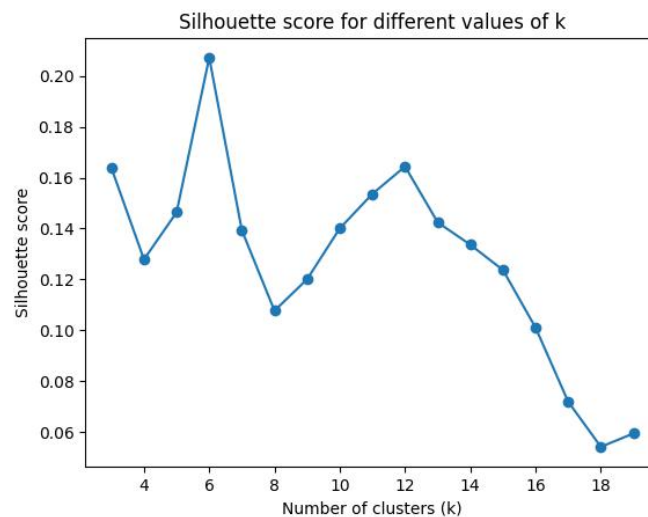


Figure 30 Silhouette score for different values of k

The clustering results are shown in a scatterplot, radar plot and column charts. These figures directly compare the distribution characteristics of each cluster in terms of rainfall, humidity, nitrogen, phosphorus, potassium and other indicators. For example, Cluster 1 shows significant advantages in terms of rainfall, humidity and nitrogen content, reflecting that its corresponding crop type may prefer a high-humidity, rainy and nitrogen-rich growing environment; Cluster 4 performs well in terms of potassium and phosphorus, suggesting that the crops in this cluster may have a higher demand for phosphorus and potassium.

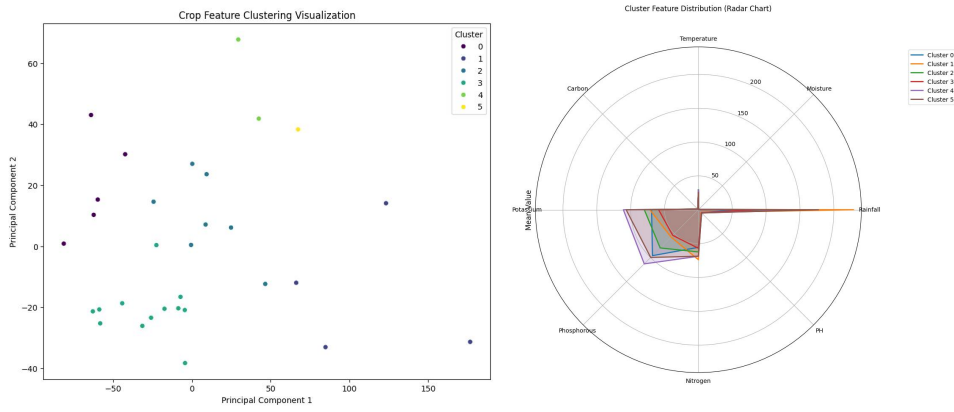


Figure 31 clusters scatter plot and radar chart

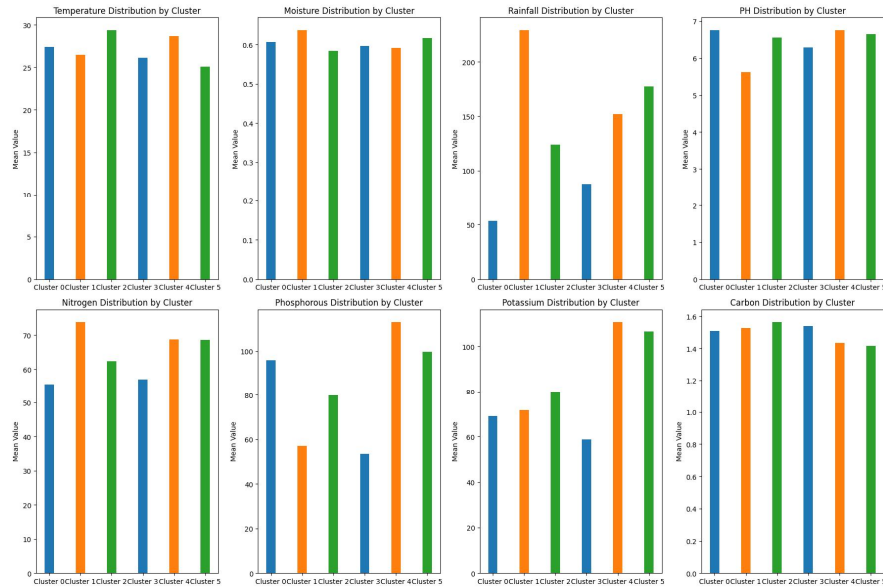


Figure 32 clusters column chart

The distribution of crops in each cluster is shown in the following. Each cluster has at least one crop and at most 12 crops.

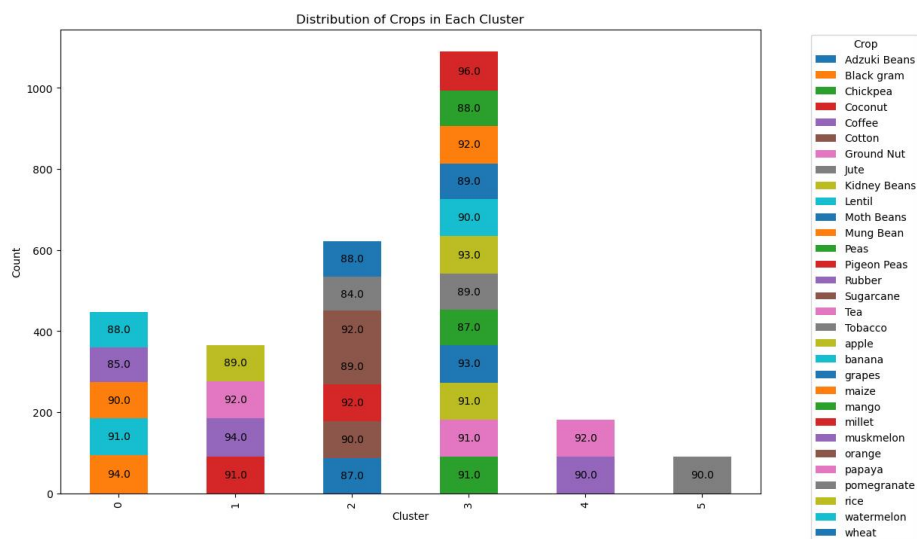


Figure 33 Distribution of Crops in Each Cluster

In future work, we can use these 6 clusters instead of the original 31 crop types as input features, effectively reducing dimensionality and improving the model's accuracy and generality.

6 Conclusion

This study highlights machine learning's transformative power in sustainable agriculture's fertilizer recommendation. Analyzing 3,100 samples of soil, climate, and crop data, we developed a framework integrating tree - based models, clustering, and imbalance - handling. The optimized Random Forest, with SMOTE and class weighting, achieved 97% accuracy and 0.957 F1 - score for minority classes, outperforming baseline models.

K - means clustering reduced crop - type dimensions from 31 to 6 interpretable clusters, simplifying features while keeping agricultural relevance. pH, nitrogen, and phosphorus were identified as key fertilizer - selection factors, in line with agronomic principles.

However, limitations persist. Minority classes are noise - sensitive due to scarce samples, and synthetic data risks overfitting. Future work should expand datasets and integrate advanced techniques. This research bridges AI and agriculture, offering a scalable solution to cut over - fertilization, lower costs, and promote sustainability, guiding smarter farming in environmental challenges.

In real-life scenarios, farmers can use our framework by focusing on soil pH, nitrogen and phosphorus to target fertiliser choices; grouping crops to simplify decision-making, where crops in a particular grouping may require fertilisers enriched in certain nutrients; and deciding on fertiliser use based on threshold alerts derived from analyses, such as the use of general-purpose fertilisers when specific conditions (e.g., humidity) are met.

Reference

[1] Database: <https://www.kaggle.com/datasets/nishchalchandel/fertilizer-recommendation>

Appendix

Questions&Answers

1.The first requirement of the project is "An objective that involves extracting knowledge from a real-world dataset."There is a lack of information regarding the source of the dataset. Please supplement where the dataset comes from and whether it is from real world.

-Danyu LYU

Reply:Here is the source website:

<https://www.kaggle.com/datasets/nishchalchandel/fertilizer-recommendation>. This dataset was created as part of the author's hackathon project, Samrudha, which focuses on AI-powered solutions for agriculture.

2.Maybe you can introduce the clustering algorithm and specific indicator values.

-Jiayu ZHENG

Reply: We have supplemented the reasons and technical details for the application of the clustering algorithm in our dataset in the cluster section in report, and explicitly given the key evaluation indicators: the highest silhouette coefficient is 0.207 when $k=6$.

3.Some of the images in the EDA section are not intuitive enough and could be streamlined and improved appropriately, but the presentation of the project process and methodology is complete.

-Ting TONG

Reply: Thanks for your feedback.We've added some new charts to better present the data models. Static images in the PPT aren't intuitive enough. The presentation in Jupyter notebook will be more interactive and intuitive.

4.Is there a possible overfitting problem? Has this issue been considered?

-Yiguo ZHOU

Reply:Yeah, we addressed the overfitting risks in the following ways. First off, we used Stratified 10 - Fold Cross-Validation, this method ensures that the model performs stably across different subsets of the data.Then, we conducted Noise Robustness Testing, we added Gaussian noise and performed random rotations. The results showed that the minority classes are particularly sensitive to these changes. This finding reminds us that we need to collect more data on those less - represented fertilizers in the future.

5.The project is really well thought out. I especially liked how you tackled data imbalance and also included cluster analysis to simplify crop categories. It shows a good balance between technical detail and practical impact for farmers. Maybe as a small idea, you could briefly mention how farmers might actually use the model's output in real scenarios.

-Xinyi CAO

Reply: Farmers can use our framework in the following ways: focus on soil pH, nitrogen, and phosphorus for targeted fertilizer selection; simplify decision - making with crops grouped into clusters, where crops in a specific cluster may need fertilizers rich in certain nutrients; and rely on threshold alerts from analysis to decide on fertilizer use, like using general - purpose fertilizer when a certain condition (e.g., moisture level) is met.

6.The PPT is done too professionally! So many data tables are organized in an orderly manner, using clear logic and intuitive comparisons to make complex information clear at a glance. The color scheme and key annotations of the table are particularly well done, and key conclusions can be grasped at a glance. With coordinated explanations, these data are no longer dry numbers, but very persuasive evidence.

-Jiajun FANG

Reply:Thanks for recognition.

7. The quality of PPT need to be improved. The introduction to the dataset should be improved since there is a lack of information of the source of the dataset. The pipeline is presented relatively clearly. Diverse methods were applied and discussion should be improved by discussing the reasons behind results in more detail.

-Kefeng ZHENG

Reply:Thank you for the feedback. In the in-depth discussion, we interpreted results such as how SMOTE affects minority classes, justify the use of K-means clustering, link noise sensitivity to data scarcity, and compare different models.

The authors analysed the factors related to agricultural cultivation, the data was pre-processed properly, the visualisation was very comprehensive, multiple views were used, in terms of model selection, the random forest achieved an accuracy of 0.97, is it a bit too optimistic, in the radar chart on page 14, the presentation of the data is not so friendly for the reader, overall it is a more successful project

-Junyi SHEN

Reply: The radar chart can intuitively display the comparison of each attribute of each cluster. Of course, we also added a cluster bar chart for each attribute in the report, which can be more friendly to readers.

8. Good research and video, but is there a lack of comparative analysis between a few models? I personally would prefer to see the reasons for the different accuracy rates between these models

-Ziyang ZHU

Reply:We compared three models. Logistic Regression (67% accuracy) struggled with skewed features and class imbalance. Decision Tree (94% accuracy) handled nonlinearity but failed on minority classes. Random Forest (97% accuracy), optimized via SMOTE and class weighting balanced accuracy and minority-class recall. Its superiority comes from ensemble design and robustness to high dimensionality.